

VARUN KUMAR KOTA

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 LinkedIn

 Github

Aspiring software engineer skilled in building scalable applications, distributed systems, and cloud-native pipelines using Python, Java, and Spark. Experienced with AWS, Azure, and full-stack development, with a strong foundation in AI/ML (PyTorch, TensorFlow) for production-ready solutions.

SKILLS

- **Programming:** Python, SQL, R, Java, JavaScript, Bash/Shell, Jira, Data Structures, Algorithms
- **Data Engineering & ETL:** Airflow, Apache Kafka, Apache Spark, Hadoop, AWS Glue, ETL Pipelines, dbt, parquet
- **AI/ML:** ML, Deep Learning, NLP, Scikit-learn, PyTorch, Tensorflow, RAG, LangGraph
- **Big Data & Storage:** Delta Lake, S3, HDFS, Data Lake Architecture, Hive
- **Cloud & Databases:** AWS (Lambda, Glue, Redshift, DynamoDB), Azure, Snowflake, Databricks, PostgreSQL
- **DevOps & CI/CD:** Git, GitLab, Jenkins, Docker
- **Data Analysis:** Power BI, Tableau, Excel, Pandas, NumPy, Matplotlib, Seaborn, Statistical, Time Series, Regression Analytics, Predictive Analysis
- **Web & APIs:** Rest API, FastAPI, Django, Streamlit, HTML5, CSS, Bootstrap, ReactJs

WORK EXPERIENCE

- AI Research Assistant** University at Buffalo Jan 2024 - May 2024
Carbon Emission Reduction Techniques [Python, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas]
- Researched GPT-2's **energy efficiency & carbon reduction** using **quantization, distillation, low-rank factorization**.
 - Optimized GPT-2 efficiency: **19.8% CO2 reduction** with only **6% perplexity increase** via quantization.
 - Achieved **45.2% emissions reduction** combining distillation and quantization, demonstrating significant carbon savings at the cost of substantial performance degradation by **57.4%**, suitable for extreme efficiency scenarios.
- Kreditbee Software Engineer** Jan 2022 – Dec 2023
- Developed and optimized UPI payment module with bank account integration, PIN management, and migration of legacy modules from Redux-Saga to React Query, improving maintainability, scalability, and overall app performance.
 - Enhanced cross-platform user experience by optimizing load performance (cutting main chunk size by 33KB) and leading the development of 130+ reusable UI components with Storybook, reducing front-end development time by 40% and ensuring design consistency across Android, iOS, and Web.
 - Built and deployed RESTful APIs (FastAPI) and designed microservices on AWS ECS and Lambda, enabling real-time reporting, scalability, and availability.
 - Implemented Kafka-based pipelines and ETL workflows, streamlining real-time event processing and secure data integration.
 - Enhanced system observability with AWS cloudWatch, Azure App Insights, improving monitoring and incident resolution.

ACADEMIC PROJECTS

- Cloud Mart: E-commerce Store** [AWS EC2, Kubernetes, Docker, CI/CD]
- Devised a microservices-based e-commerce solution using Kubernetes and Terraform, delivering fault-tolerant infrastructure with 95% uptime.
 - Architected a microservices based infrastructure with containerized services, integrating AI-driven customer interactions, resulting in a 30% increase in scalability and 20% faster operational efficiency.
- Enterprise-Grade Full-Stack Application with Kubernetes**
- Architected and developed a robust microservices application using Spring Boot, implementing an API Gateway, Kafka for asynchronous communication, and a polyglot persistence strategy with both MongoDB and MySQL.
 - Engineered a scalable deployment workflow using Docker and Kubernetes for container orchestration, while leveraging the Grafana Stack (Prometheus, Loki, Tempo) for end-to-end monitoring, logging, and tracing.
- Medical AI Assistant**
- Engineered an end-to-end RAG pipeline using Gemini 2.5 Flash for advanced text processing and summarization, and a Pinecone vector database for scalable semantic search across 4K medical records.
 - Created a Streamlit chatbot with an enhanced retriever that combines semantic and keyword-based scoring for highly accurate responses, while also incorporating a query classifier for user safety.
- Enterprise Data Engineering Workflow on Azure Cloud**
- Built an end-to-end data pipeline using Azure Data Factory, Data Lake Gen2, Databricks (PySpark), Synapse, and Power BI, improving data availability by 85% and reducing reporting lag by 70% through real-time processing and optimized transformations.
 - Automated data ingestion, transformation, and reporting workflows, cutting manual effort by 90% and data processing time by 60%, while enabling business insights through interactive Power BI dashboards integrated with Synapse external tables.

EDUCATION

- University at Buffalo, The State University of New York** Jan 2024 – May 2025
Master of Science, Data Science, CGPA: 3.4/4
Buffalo, New York
- Vellore Institute of Technology** July 2019 – May 2023
Bachelor of Technology, Electronics and Computer Engineering, CGPA: 8.86/10
Chennai, India

CERTIFICATIONS

- AWS Certified Cloud Practitioner, Data Engineer - Associate